

Peer Review

Review of: "Accuracy Without Profit: A Statistical Evaluation of Machine Learning Profitability in the English Premier League"

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This paper studies whether machine learning models can make consistent profits in the English Premier League betting market. The main conclusion is interesting: even when the models achieve reasonable prediction performance, it is still difficult to obtain stable returns because of alpha decay and calibration error. I think this is a meaningful topic, and the paper is easy to follow overall.

I appreciate that the authors use the Diebold-Mariano test to examine whether the model predictions are statistically different from the market odds. I also appreciate that the code is publicly available, which improves transparency and reproducibility.

I have several suggestions that could strengthen the paper.

First, the compared models are mostly tree-based methods. It may be helpful to include an additional benchmark, such as a neural network model, to show whether the conclusions are specific to tree models or more general.

Second, the training data expands year by year. Because the paper discusses market change over time, it would be interesting to compare this setting with a rolling-window approach, for example, training only on the most recent five years. This may also help show whether feature importance changes over time.

Third, the paper should include basic descriptive statistics of the dataset, such as the number of matches and the proportions of home wins, draws, and away wins. This would help readers better understand the prediction task.

Fourth, the description of the simulation is not detailed enough. It is not clear what exactly was varied or resampled in the 20,000 simulation runs. More explanation is needed for reproducibility.

Fifth, the paper argues that calibration is very important. In that case, it would be useful to perform an actual calibration step using past training data or validation data, and then compare the calibrated and uncalibrated results on the test data.

Finally, to better understand the role of the L5 features, I would like to see an ablation study: odds-only, L5-only, and odds-plus-L5 models. This would clarify how much additional value the L5 features provide.

Overall, I think the paper addresses an interesting and practical question, and it has good points such as transparency and a clear experimental setup. At the same time, the paper would be stronger with more detailed analysis, a clearer simulation description, and additional comparison experiments.

Declarations

Potential competing interests: No potential competing interests to declare.